

Integration WS 9

Evaluate Q1-12

1. $\int_0^{\frac{\pi}{3}} \frac{1}{1+\sin x} dx$

2. $\int_0^{\frac{\pi}{2}} \frac{dx}{2+\cos x}$

3. $\int \frac{dx}{\sin x - \cos x}$

4. $\int_0^{\frac{\pi}{2}} \frac{1}{3+5 \cos x} dx$

5. $\int \frac{1}{x^2 \sqrt{x-1}} dx$

6. $\int \frac{2x^2-x+20}{(x-2)(x^2+9)} dx$

7. $\int \frac{12}{(x^2+4)(x^2+16)} dx$

8. $\int_0^{\frac{\pi}{4}} \frac{1-\tan x}{1+\tan x} dx$

9. $\int_0^{\frac{\pi}{2}} \sqrt{1+\sin 2x} dx$

10. $\int \frac{dx}{x^2 \sqrt{9-x^2}}$

11. $\int \frac{x^2 dx}{\sqrt{16-x^2}}$

12. $\int \frac{\sqrt{4-x^2}}{x^2} dx$

13. Let $a > 1$. Find the value of $\int_0^{\pi} \frac{1}{a-\cos \theta} d\theta$

14. Let $I_k = \int \cos^k x dx$ and $J_n = \int \frac{1}{(x^2+a^2)^n} dx$.

a. Show that $I_k = \frac{1}{k} \sin x \cos^{k-1} x + \frac{k-1}{k} \int \cos^{k-2} x dx$.

b. Using a suitable substitution, show that

$$J_n = \frac{1}{a^{2n-1}} \int \cos^{2(n-1)} \theta d\theta.$$

c. Hence or otherwise, give the exact value

$$\text{of } \int_0^{\frac{1}{2}} \frac{1}{\left(x^2+\frac{1}{4}\right)^3} dx.$$

15. Find a formula finding I_n in terms of I_{n-1} :

a. $I_n = \int x^n e^x dx$

b. $I_n = \int x^n e^{-2x} dx$

c. $I_n = \int (\log_e x)^n dx$

d. $I_n = \int \frac{dx}{(x^2+a^2)^n}$

16. $I_n = \int \cot^n x dx$

a. Show $I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$

b. Hence find $\int \cot^6 x dx$

17. $I_n = \int \cos^n x dx$

a. Show $I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$

b. Hence find $\int \cos^6 x dx$

c. Hence find $\int_0^{\frac{\pi}{2}} \cos^8 x dx$

18. $I_n = \int \sin^n x dx$

a. Show $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$

b. Hence find $\int_0^{\frac{\pi}{2}} \sin^8 x dx$

c. Find $\int_0^{\frac{\pi}{4}} \sin^4 x dx$

19. $I_n = \int \tan^n x dx$

a. Show $I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$

b. Hence find $\int \tan^5 x dx$

c. Evaluate $\int_0^{\frac{\pi}{4}} \tan^8 x dx$

20. $I_n = \int (\sin^{-1} x)^n dx$

a. Show $I_n = x(\sin^{-1} x)^n + n\sqrt{1-x^2}(\sin^{-1} x)^{n-1} - n(n-1)I_{n-2}$

b. Hence find $\int_0^1 (\sin^{-1} x)^3 dx$

21. Show that $U_n = -\frac{2n}{2n+3} U_{n-1}$, if $U_n =$

$$\int_{-1}^0 x^n (1+x)^{\frac{1}{2}} dx, \text{ hence evaluate } U_3.$$

22. If $I_n = \int e^{mx} \tan^n x dx$, show that $I_n =$

$$\frac{e^{mx}}{n-1} \tan^{n-1} x - \frac{m}{n-1} \cdot I_{n-1} - I_{n-2}$$

23. Find $\int x^4 e^x dx$, using the reduction formula.

24. If $I_n = \int \sec^n x dx$, prove that $I_n =$

$$\frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \cdot I_{n-2}.$$

25. Prove: $\int x^n e^{-x} dx = -e^{-x} x^n +$

$$n \int x^{n-1} e^{-x} dx. \text{ Hence, find : } \int x^3 e^{-x} dx.$$

26. Show that $\int_0^{2\pi} e^x \cos nx dx = \frac{1}{n^2+1} (e^{2\pi} -$

$$1). \text{ Hence evaluate } \int_0^{2\pi} e^x \cos 3x \cos x dx \text{ in simplest exact form}$$

27. Given that $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^n x dx$, for $n = 1, 2, \dots$

- a. Show that $I_1 = \frac{1}{2} \ln 2$
- b. Show that $I_{n-2} + I_n = \frac{1}{n-1} \left(3^{\frac{1}{2}(n-1)} - 1 \right)$
for $n = 2, 3, 4 \dots$
- c. Find I_5

28. For each integer $n \geq 0$

let $I_n = \int_0^1 x(x^2 - 1)^n dx$

- a. Using the method of integration by parts,
show that for $n \geq 1$, $I_n = -\frac{n}{n+1} I_{n-1}$
- b. Hence, or otherwise, show that for $n \geq 0$,
 $I_n = \frac{(-1)^n}{2(n+1)}$
- c. Show that for n is odd that $I_n < I_{n+2}$ for
all $n \geq 0$
- d. Deduce that $-\frac{1}{4n} < I_{2n+1} < -\frac{1}{4(n+2)}$

29. a. Given that $\sin x \geq \frac{2x}{\pi}$ for $0 < x < \frac{\pi}{2}$, explain

why $\int_0^{\frac{\pi}{2}} e^{-\sin x} dx < \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx$

b. Show that $\int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx = \int_0^{\frac{\pi}{2}} e^{-\sin x} dx$

c. Hence, show that $\int_0^{\pi} e^{-\sin x} dx < \frac{\pi}{e} (e - 1)$

30. It is given that if $J_n = \int \cos^{n-1} x \sin nx dx$ and

$n \geq 1$, then $J_n = \frac{1}{2n-1} [(n-1)J_{n-1} -$

$\cos^{n-1} x \cos nx]$

Use this reduction formula to show that:

$$\int_0^{\frac{\pi}{4}} \cos^2 x \sin 3x dx = \frac{1}{60} (28 - \sqrt{2})$$

31. By using the substitution $y = \frac{\pi}{2} - x$, show

that $I = \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+\cot x} dx$. Hence
find the value of I .

32. If a function $f(x)$ is continuous for $a \leq x \leq b$

a. Show that $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

b. Hence prove that $\left| \int_0^{\pi} 4^x \cos x dx \right| \leq \frac{2^{2\pi}-1}{2 \ln 2}$

33. a. Use the substitution $x = a - t$ to prove that

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

b. Hence evaluate $\int_0^{\frac{\pi}{2}} \ln(\tan x) dx$

34. a. Given that $I_{n,a} = \int \frac{dx}{(x^2+a^2)^{n+\frac{1}{2}}}$, show that

$$I_{n,a} = \frac{1}{a^{2n}} \int \cos^{2n} u du$$

b. Hence, find $I_{2,1}$ leaving your answer in
terms of x .

35. Show that $\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{n-1}{n} \int_0^{\frac{\pi}{2}} \sin^{n-2} \theta d\theta$,

and hence show that $\int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta = \frac{3\pi}{16}$

36. Apply successively the result of Q35 to show

that $\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{(n-1)(n-3)\dots 4.2}{n(n-2)\dots 3.1}$ if n is odd

and $\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{(n-1)(n-3)\dots 3.1}{(n(n-2)\dots 4.2)} \cdot \frac{\pi}{2}$ if n is

even.

Answer

1. $\sqrt{3} - 1$

2. $\frac{\pi}{3} \sqrt{3}$

3. $\frac{1}{\sqrt{2}} \ln \left[\frac{t+1-\sqrt{2}}{t+1+\sqrt{2}} \right], t = \tan \left[\frac{x}{2} \right]$

4. $\frac{1}{4} \ln 3$

5. $\tan^{-1} \sqrt{x-1} + \frac{1}{x} \sqrt{x-1}$

6. $2 \ln|x-2| - \frac{1}{3} \tan^{-1} \frac{x}{3}$

7. $\frac{1}{2} \tan^{-1} \frac{x}{2} - \frac{1}{4} \tan^{-1} \frac{x}{4}$

8. $\frac{1}{2} \ln 2$

9. 2

10. $\ln(x + \sqrt{x^2 + 4})$

11. $8 \sin^{-1} \left(\frac{x}{4} \right) - \frac{x}{2} \sqrt{16 - x^2}$

12. $-\frac{\sqrt{4-x^2}}{x} - \sin^{-1} \frac{x}{2}$

13. $\frac{\pi}{\sqrt{a^2-1}}$

14. $c. 8 + 3\pi$

15. a. $e^x x^n - n I_{n-1}$;

b. $-\frac{1}{2} x^n e^{-2x} + \frac{n}{2} I_{n-1}$;

c. $x (\ln x)^n - n I_{n-1}$;

d. $\frac{1}{a^2(2n-2)} \left[\frac{x}{(x^2+a^2)^{n-1}} + (2n-3) I_{n-1} \right]$

16. b. $-\frac{\cot^5 x}{5} + \frac{\cot^3 x}{3} - \cot x - x + C$

17. b. $\frac{\cos^5 x \sin x}{6} + \frac{5 \cos^3 x \sin x}{24} + \frac{15 \cos x \sin x}{48} + \frac{15x}{48} + C$; c. $\frac{32\pi}{256}$

18. b. $\frac{35\pi}{256}$; c. $\frac{3\pi}{32} - \frac{1}{4}$

19. b. $\frac{\tan^4 x}{4} - \frac{\tan^2 x}{2} - \ln \cos x + C$;

c. $\frac{\pi}{4} - \frac{76}{105}$

20. b. $\frac{\pi^3}{8} - 3\pi + 6$

21. $-\frac{32}{256}$

23. $e^x(x^4 + 4x^3 + 12x^2 - 24x + 24)$

24. $\frac{28}{15}$

25. $-e^{-x}(x^3 + 3x^2 + 6x + 6)$

26. $\frac{11}{85}(e^{2\pi} - 1)$

27. $I_5 = 1 + \frac{1}{2} \ln 2$

31. $\frac{\pi}{4}$

33. 0

34. b. $\frac{x(3+2x^2)}{3(1+x^2)\sqrt{1+x^2}} + C$